

FakeBand: Virtual Band Music Performance with Balanced Interface for Individual Score/Rhythm Play and Inter-player Expression Coordination

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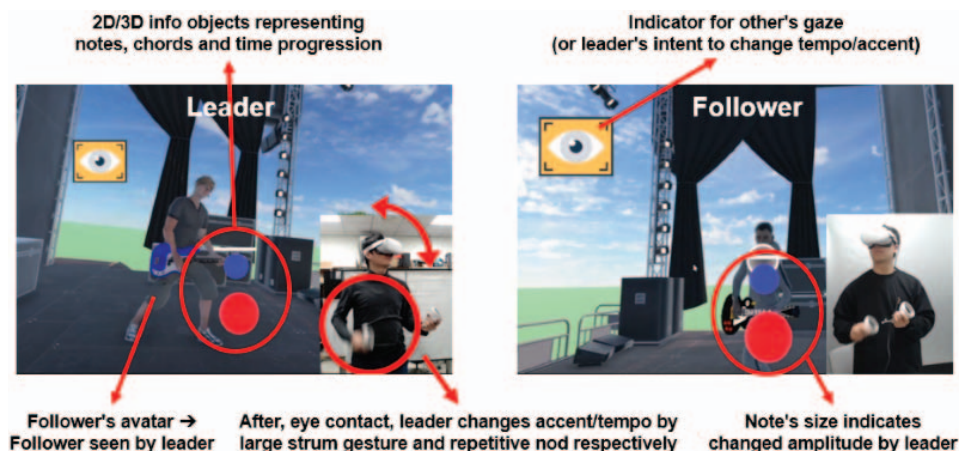
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Figure 1: Typical two player FakeBand scene and its interface. FakeBand is composed of (1) a stage scene with a music video backdrop, (2) two instrument (bass and lead guitar) playing avatars (a leader and a follower) representing the users, (3) 2D/3D info objects representing bars, measures, chords, and notes for the respective part which appear in synch to the on-going music, (4) hand/finger based guitar-like interface for the user for the rhythm and chord play, and (5) body based interface to signal and change tempo and accent. The user needs to press the button with one's right hand/finger installed on the guitar-like interface when the notes reach near the timeline. If rightly pressed, the bass note is played. The user also has to use the controller with one's left hand to change the chord accordingly (by simple slide gesture, either up or downward). The tempo and accent change is signaled by the leader's gaze caught by the follower. The actual tempo is changed by the leader with repetitive nods and the accent with large arm motions.

ABSTRACT

In this poster, we feature a two-user immersive VR music performance system called the "FakeBand". It resembles the rhythm game for a guitar play-along, but extended in its interaction model and interface design, focusing on the band performance experience in terms of coordination and collective musical expression. Coordinating timings for the tempo/accent change begins by the leader and the follower exchanging glances and using bodily gestures - nodding for tempo and large arm motion for accents. The interaction model and separated interface between the rhythm/chord play and music coordination were important in bringing about the active and frustration-free musical experience.

Index Terms: Virtual reality (VR); Human-computer music performance; Virtual human; Multimodal interaction;

1 INTRODUCTION

Music is one of the essential human activities and forms of enjoyment. Active music performance in particular offers another

dimension in music enjoyment beyond just listening to it. However, in many cases, music playing is constrained by the difficulty to find the right space and summon the people to play with. This is especially true for playing instruments that have the supportive roles such as the bass guitar, drum and brass. For this, while automatic accompaniment systems have been developed [2], the user still needs to be proficient in the instrument of one's part which usually requires years of practice. Furthermore to play a given piece without any error and express one's emotion, even more practice will be needed to a level that the piece is almost memorized, only so that the player will have sufficient cognitive room to minutely control the music to one's liking.

Rhythm games, such as the Guitar Hero [1], partly addresses this issue in the form of a game. It allows non-players to act and feel as if one can play (e.g. the guitar by showing the rhythmic notes/events along the music play and lead the user to act on them by simple timely button press, partly eliminating the aforementioned "skill/practice" problem). This type of a game has found good success and even though "reactive" in nature, it partly gives the false sense of control to the user. Nevertheless, the extent of the active music play is limited, because even for moderately difficult pieces, the notes to react become large in number and their timings complicated, and the game starts to become more like a busy physical reflex testing game. On top of this, there is no interface to control or coordinate the expression, nor does the user have any mental room to make such control.

In this poster/demonstration, we feature a two user immersive

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VR music performance system called the "FakeBand". It adopts a hand/finger rhythm play like interaction to free the novices from making mistakes and worrying about them. Furthermore, non-occupied body parts are used to make coordinated play, namely, tempo and accent change. This balanced and managed workload and physically separated interface (hand/finger vs. other body parts) is the key to delivering the band-like jam making experience difficult to attain as novice players.

2 FAKEBAND

2.1 System and Scene

FakeBand is designed as a two-user (for now) immersive virtual reality system to give the sense of place (e.g. stage) and social presence (e.g. co-playing band members) as such elements can elevate the musical experience, especially that of a band. The scene is composed of a stage with a music video backdrop and the two guitar playing avatars (a leader and a follower) representing the users. There is also system/AI controlled guitar playing avatar (band member) and a crowd reacting to the music play (see Figure 1).

2.2 Note Level Rhythm and Chord Change

The rhythmic note and chord playing are accomplished by two controller-based guitar-like input interface (see Figure 1): (1) the right hand/finger supplying the button press for the note level rhythm progression, and (2) left hand making slide action to change the chord. 2D/3D graphic musical info (extracted by an off-line music analysis process) objects representing bars, measures, chords, and notes appear in synch to the on-going music similarly to the rhythm game visualization. With such guides, the user needs to press the button with one's right hand/finger when the notes reach near the timeline. If rightly pressed, the bass note is played.

Note that the user only has to input the notes and change the chord with the right timing, and does not "select" a particular key or chord. Such an interaction model was chosen deliberately to minimize the skill as required from the novice player, so that, given that the user is already familiar with the rhythm of the song, one would have almost no problem in progressing one's part, and thereby will have the mental capacity to pay attention to other things like musical expression with regards to tempo and accent, and looking around and enjoying the music with other members of the band and crowd. Note that the difficulty level can be adjusted to the point where the user can freely improvise one's rhythm or chord.

2.3 Coordinated Tempo and Accent Control

The two main expression parameters in music are the "tempo" and "accent (or amplitude)". There are many musical notations in the score that refer to such how players should control these. However, they are also often spontaneously adjusted while playing. Furthermore, in a band (or orchestral/chamber music) play, expression control should be coordinated with other players. The interface for the expression coordination is purposely designed to be body-based as musical expression is often conveyed by bodily "groovy" gestures. Non-verbal interaction is needed anyway since music is being played as well. The player's hands are occupied anyway for the note/chord play.

Maezawa and Yamamoto [3] explored the "nudging" interface for a real-virtual coordinated chamber music performance. Nudge is a one way signal, an exaggerated act by the leader to indicate a start of a phrase. Here, we extended this idea and devised a two-way cue exchange interaction for tempo and accent coordination. The interface operates by the designated leader (among the two) first looking at the other player (follower's head) which is easily detected by tracking the view direction of the partaking user. When the leader wishes to signal an event and looks at the follower, a graphical indicator appears to the follower to make sure the follower responds and make mutual eye contact (yellow eye shaped icon in

Figure 1). Once the eye contact is made, the tempo can be changed by the leader by one's nodding gesture either fast or slow. Through such signaling and acknowledgement, the follower is not startled and can readily adjust to the changed tempo. The same signaling and acknowledgement process is used for accent indication as well. Once the mutual eye contact is made, the leader can give accents to one's note playing accompanied by large arm motion, which itself is visible to the follower and also visualized on the follower's side as "enlarged and emphasized" note events. While the eye contact may not be absolutely necessary to coordinate or make indications of various music expression, it promotes the sense of lively human-to-human communication.

3 SUBJECTIVE PILOT EVALUATION

We conducted a preliminary pilot subjective evaluation test with 5 participants. All participants were males, aged between 23 to 26 ($M=24.4$, $SD=1.02$), and had more than 20 hours of VR experience. In terms of the musical experience, 4 participants had musical/instrument training of at least two years. None had any band experience. The users expressed that even with just the simple rhythm based play, handling the chord change and tempo/accents coordination at the same time were still challenging, yet resulted in a more satisfying musical experience (e.g. than the conventional rhythm games). While the body-based musical coordination may have been inspired from the real band playing practice, note that the tested users did not really have such actual experience. Like any unfamiliar interaction and interface, some minimal training would be necessary: most users were confident they would be able to enjoy the virtual band playing much more in time. Few users complained of the passiveness on the follower, having to just react to the leader, and inaccuracy of the eye contact. In the future, we plan to conduct a more formal comparative user experience evaluation, and also explore an interaction model of a two/multi-way negotiation style coordination as well.

4 CONCLUSION

In this work, we showcased "FakeBand" which allows the two users to, with minimal skills, experience band playing by active control of the rhythm/chord progression and tempo/accents coordination among the users. The interface is designed to support the rhythm/chord playing and expression control at the same time with the balanced difficulty level. This made the users to concentrate more on the "music" itself rather than worrying about hitting the wrong notes and rhythm.

We believe that this interaction design principle carries over to other domains that requires multi-faceted control - where the main task should be as easy as possible (perhaps carried out by the main limbs/body parts), and the rest of the body be employed for supplementary/affective/coordinating aspect. Good examples would be team sports and group collaboration tasks. For this, the system needs to be extended not only to support more than just two players, but also multi-party (more than two) and more accurate non-contact/verbal coordination.

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